

Solution of Exercise 2

Exercise 2.1 Simplex method

a) Check for basic feasible solution

Basis $\mathcal{B}$	Basis matrix $\mathbf{B}$	$\mathbf{x}_B = \mathbf{B}^{-1}\mathbf{b}$	$\mathbf{x}_N = \mathbf{0}$	basic feas. sol.?
$\mathcal{B}_1 = \{1, 3\}$	$\mathbf{B} = \begin{bmatrix} 2 & 1 \\ 2 & -3 \end{bmatrix}$	$x_1 = 15/16 > 0$ $x_3 = 1/8 > 0$	$x_2 = 0$ $x_4 = 0$	yes
$\mathcal{B}_2 = \{1, 4\}$	$\mathbf{B} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$	-	-	no, since $\mathbf{B}$ singular
<u>Add-On:</u> $\mathcal{B}_3 = \{2, 3\}$	$\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 0 & -3 \end{bmatrix}$	$x_2 = 2.5 > 0$ $x_3 = -0.5 < 0$	$x_1 = 0$ $x_4 = 0$	no

b) Solution with simplex method and start basis $\mathcal{B}_1 = \{1, 3\}$

$$\mathcal{B} = \{1, 3\}, \quad \mathbf{B} = \begin{bmatrix} 2 & 1 \\ 2 & -3 \end{bmatrix}, \quad \mathbf{x}_B = \begin{bmatrix} 15/16 \\ 1/8 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\mathcal{N} = \{2, 4\}, \quad \mathbf{N} = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} 2 \\ 0.5 \end{bmatrix}$$

Evaluation of the KKT conditions

$$\boldsymbol{\lambda} = -(\mathbf{B}^T)^{-1}\mathbf{c}_B = \begin{bmatrix} -9/8 \\ 5/8 \end{bmatrix}, \quad \boldsymbol{\mu}_N = \mathbf{c}_N + \mathbf{N}^T\boldsymbol{\lambda} = \begin{bmatrix} 7/8 \\ -0.5 \end{bmatrix}$$

shows, that this point is not the minimum ($\mu_{N,2} = \mu_4 < 0$).

One step of the simplex algorithm leads to

$$s = 4, \quad \mu_s = -0.5, \quad \mathbf{d} = \mathbf{B}^{-1}\mathbf{A}_4 = \begin{bmatrix} 3/8 & 1/8 \\ 2/8 & -2/8 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$r = 1, \quad x_s^+ = \frac{x_{B,1}}{d_1} = 15/16$$

$$\mathbf{x}_B^+ = \mathbf{x}_B - \mathbf{d}x_s^+ = \begin{bmatrix} 0 \\ 1/8 \end{bmatrix} = \begin{bmatrix} x_1^+ \\ x_3^+ \end{bmatrix}, \quad \mathbf{x}_N^+ = \begin{bmatrix} 0 \\ x_s^+ \end{bmatrix} = \begin{bmatrix} 0 \\ 15/16 \end{bmatrix} = \begin{bmatrix} x_2^+ \\ x_4^+ \end{bmatrix}.$$

For the next simplex step, the elements from $\mathcal{B}$ and $\mathcal{N}$ are interchanged with the indices $r \in \mathcal{B}$ and $s \in \mathcal{N}$, resulting in the *new feasible solution*

$$\mathcal{B} = \{3, 4\}, \quad \mathbf{B} = \begin{bmatrix} 1 & 2 \\ -3 & 2 \end{bmatrix}, \quad \mathbf{x}_B = \begin{bmatrix} 1/8 \\ 15/16 \end{bmatrix}, \quad \mathbf{c}_B = \begin{bmatrix} 3 \\ 0.5 \end{bmatrix}$$

$$\mathcal{N} = \{1, 2\}, \quad \mathbf{N} = \begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}, \quad \mathbf{x}_N = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \mathbf{c}_N = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Evaluation of the KKT conditions for the new point results in

$$\boldsymbol{\lambda} = -(\mathbf{B}^T)^{-1} \mathbf{c}_B = \begin{bmatrix} -15/16 \\ 11/16 \end{bmatrix}, \quad \boldsymbol{\mu}_N = \mathbf{c}_N + \mathbf{N}^T \boldsymbol{\lambda} = \begin{bmatrix} 0.5 \\ 17/16 \end{bmatrix} > \mathbf{0}.$$

Since both Lagrange multipliers $\boldsymbol{\mu}_N = [\mu_{N,1}, \mu_{N,2}]^T = [\mu_1, \mu_2]^T$ are positive, the point $\mathbf{x}^* = [0, 0, 1/8, 15/16]^T$ is the *optimal solution*.

Exercise 2.2 Active set method

a) Gradient and Hessian of the quadratic cost function:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 2(x_1 - 2.5) \\ 2(x_2 - 3.5) \end{bmatrix} \quad (1)$$

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \quad (2)$$

Checking for definiteness with Theorem 2.2 in the script:

$$\mathbf{p}^T \nabla^2 f(\mathbf{x}) \mathbf{p} = 2p_1^2 + 2p_2^2 > 0 \quad \forall \mathbf{p} \neq \mathbf{0} \quad (3)$$

$\Rightarrow \nabla^2 f(\mathbf{x})$ is positive definite, thus the function f is strictly convex.

$\Rightarrow$ This is a convex problem because the cost function is convex, and the admissible set is also convex (see Figure 2).

b) The KKT conditions are obtained using the Lagrangian function

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\mu}) = (x_1 - 2.5)^2 + (x_2 - 3.5)^2 + \boldsymbol{\mu}^T (\mathbf{C}\mathbf{x} - \mathbf{d}) \quad (4)$$

with $\mathbf{C} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \\ -2 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}$ and $\mathbf{d} = [6, -0.5, 1.5, 0, 0]^T$ resulting in

$$\begin{bmatrix} 2(x_1^* - 2.5) \\ 2(x_2^* - 3.5) \end{bmatrix} + \begin{bmatrix} 2 & -1 & -2 & -1 & 0 \\ 1 & -1 & 1 & 0 & -1 \end{bmatrix} \boldsymbol{\mu}^* = \mathbf{0} \quad (5a)$$

$$\mathbf{C}\mathbf{x}^* - \mathbf{d} \leq \mathbf{0} \quad (5b)$$

$$\boldsymbol{\mu}^* \geq \mathbf{0} \quad (5c)$$

$$\boldsymbol{\mu}^{*T} (\mathbf{C}\mathbf{x}^* - \mathbf{d}) = 0. \quad (5d)$$

c) General procedure of the active set method:

- (1) Minimize the equality-constrained problem with the currently active constraints. If a new constraint is reached in the process, it is added to the current working set.
- (2) At the minimum of the current equality-constrained problem, check the KKT constraints, i.e., compute $\boldsymbol{\mu}$ from (5a). If $\boldsymbol{\mu} \geq \mathbf{0}$ holds, the minimum of the original problem is found, if not, the constraint with the most negative μ is selected and removed from the working set (and step (1) is repeated).

The steps for the given problem are:

- $\mathbf{x}^0 = [0.5, 0]^T \Rightarrow \mathcal{A} = \{2, 5\}$. $\mathbf{x}_0$ is already the minimum with the current active constraints. Calculating the multipliers gives $\mu_2 = -4$ and $\mu_5 = -3$, i.e. let go of constraint 2 $\Rightarrow \mathcal{A} = \{5\}$.
 - $\mathbf{x}^1 = [2.5, 0]^T$ (derived graphically). Calculating multipliers gives $\mu_5 = -7$, i.e. let go of constraint 5 $\Rightarrow \mathcal{A} = \{\}$.
 - $\mathbf{x}^2 = [2.5, 1]^T$ (derived graphically). Constraint 1 becomes active $\Rightarrow \mathcal{A} = \{1\}$.
 - $\mathbf{x}^3 = [1.5, 3]^T$ (derived graphically). Calculating the multipliers gives $\mu_1 = 1$, i.e. $\mathbf{x}^3 = \mathbf{x}^*$.
- d) The multiplier μ^* measures how sensitive the optimal objective value $f(\mathbf{x}^*)$ is to a small relaxation or tightening of the active constraint.

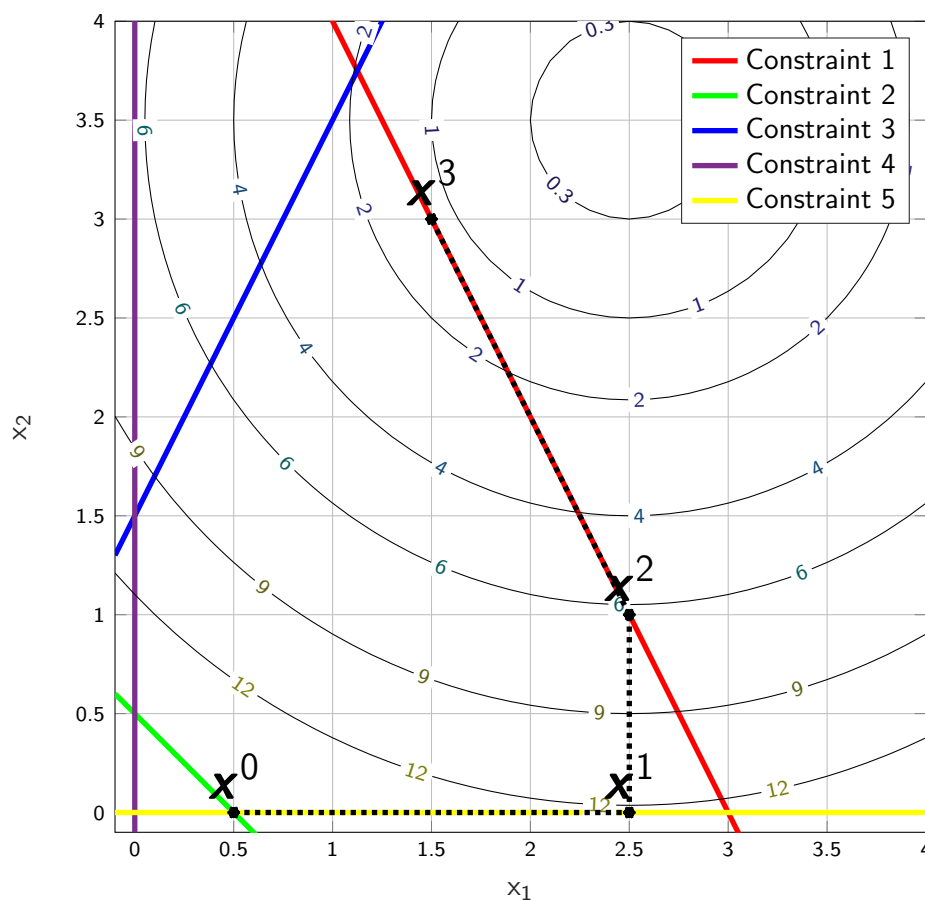


Figure 1: Iterations of the active set method.